DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& CS/IT

Calculus and Optimization

Functions and Graphs

Lecture No. 01



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Topics to be Covered



Topic

FUNCTIONS & THEIR GRAPHS

CS/IT: 7-8 Marks.

Maths

DA → 43 Marks

(2025)

1. Linear Algebra
   - Part 1 (for Both DA & CS)
   - Part 2 (only for DA)
2. Calculus
3. Prob & Stats
   - Part 1 (for Both DA & CS)
   - Part 2 (only for DA)

Function:- $f(x)$ is said to be a func^n from A to B if

for every $x \in A$, there exist unique $y \in B$ s.t $f(x) = y$

i.e $\forall x \in A$, $\exists$ unique $y \in B$ s.t $y = f(x)$

where x = Independent Variable $\simeq$ Input

& y = Dependent Variable $\simeq$ output

Def^n of func^n in My Language:-

" It is a Relationship b/n Input (x) & output (y) under certain Rules. "

& these Rules are called Domain of func^n.

Domain of $y=f(x)$ $\Rightarrow$ Set of permissible values of x is called Domain

Range of $y=f(x)$ $\Rightarrow$ Set of permissible values of y is called Range

i.e Restrictions imposed on Inputs (x) is called Domain

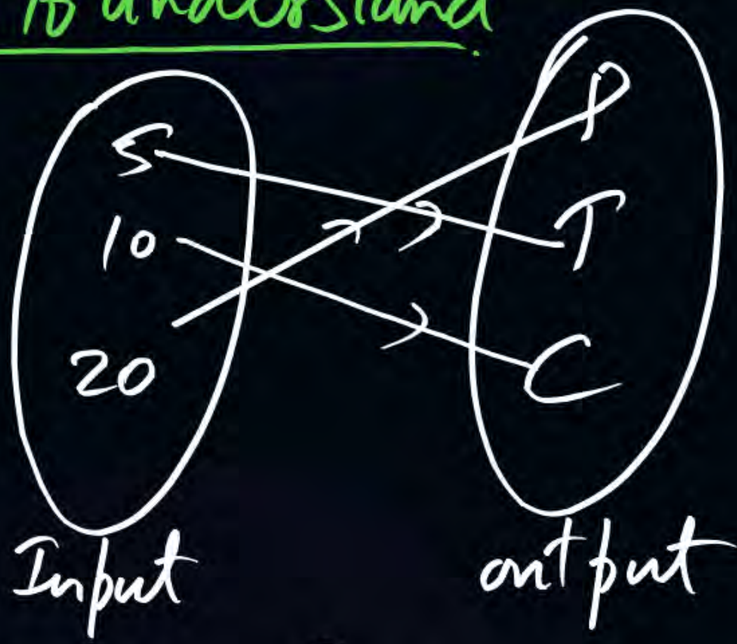
& " " " outputs (y) " " Range.

Note: **Vertical line Test** → If any Random line $\perp^{r}$ to y axis, cuts the graph only at one point, then it is a func^{n}.

& if this line cuts the graph at more than one point, then it is not a func^{n}.

Examples to understand the func^n

(1)



Input output

Domain = {5, 10, 20}

Range = {P, T, C}

Codomain = {P, T, C}

ONTO

(2)



D = {5, 10, 20}

R = {P, T}

Codomain = {P, T, C}

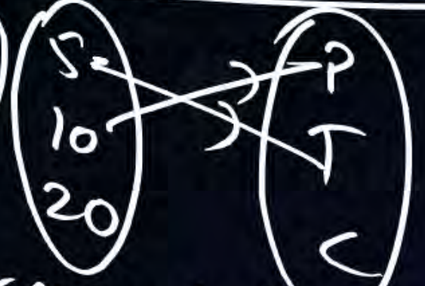
INTO

(3)



uniqueness property does not hold so it is not func^n

(4)

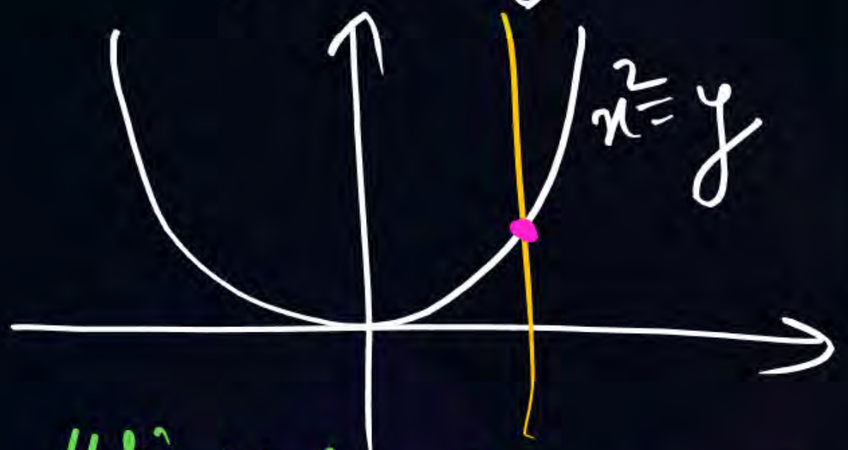


for every property does not hold so it is not a func^n.

(5) $y = x^2$ is a func^n But $y^2 = x$ is not a func^n.

At $x=4$, $y=(4)^2=16$

i.e y is unique



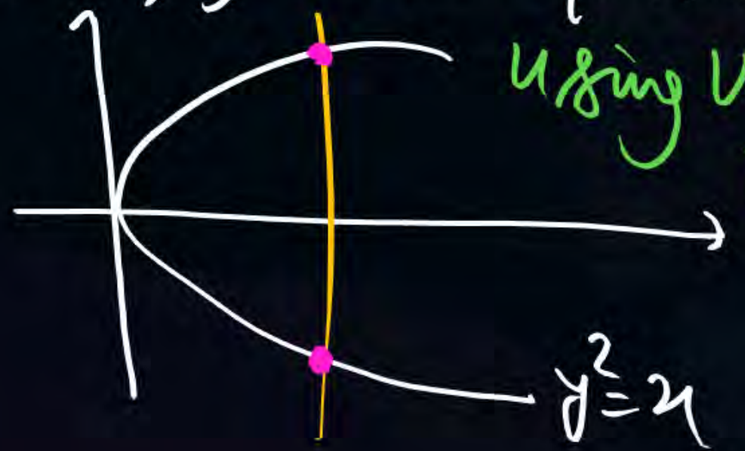
Using V-line test, it is func^n.

At $x=4$, $y^2=4 \Rightarrow y=\pm\sqrt{4}=\pm 2$

i.e at $x=4$ $\begin{cases} y=2 \\ y=-2 \end{cases}$

i.e

At $x=4$, y is not unique so it is not a func^n.



Using V-line test it is not a func^n.

Even func^n: if $f(-x)=f(x) \Rightarrow f(x)$ is called an Even func^n
& it's Graph is Symmetrical about Y axis

odd func^n if $f(-x)=-f(x) \Rightarrow f(x)$ is called an odd func^n.
& it's Graph is Symmetrical about origin i.e (I $\leftrightarrow$ III & II $\leftrightarrow$ IV)

NENO func^n if $f(-x) \neq f(x)$ & $f(-x) \neq -f(x)$ } then $f(x)$ is called NENO func^n.

it's Graph is neither Symmetrical about Y axis, nor about origin.

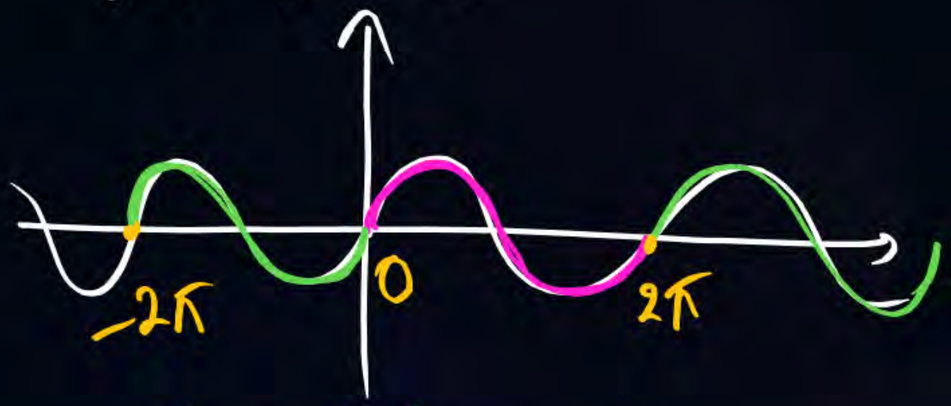
Periodic func^n → $f(x)$ is called Periodic func^n of period T if

$$\boxed{f(x+T)=f(x)}$$

for eg $\sin(x+2\pi)=\sin x$ so Period of $\sin x$ is 2π

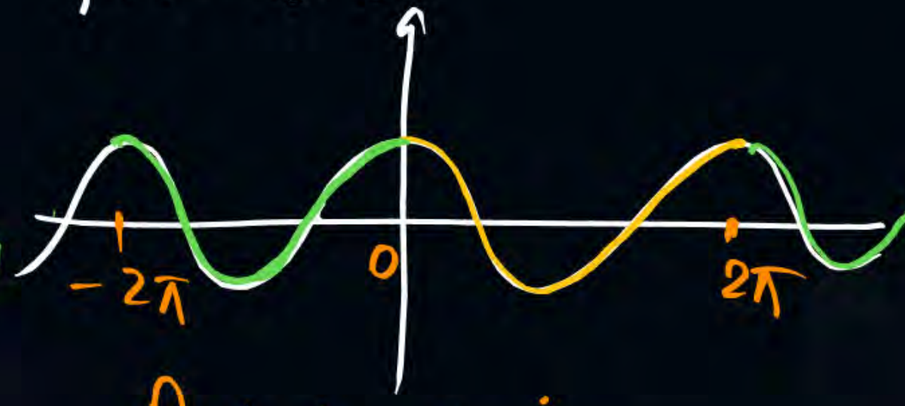
& $\tan(x+\pi)=\tan x$ i.e Period of $\tan x$ is π

$f(x)=\sin x$



Period $(T)=2\pi$

$f(x)=\cos x$

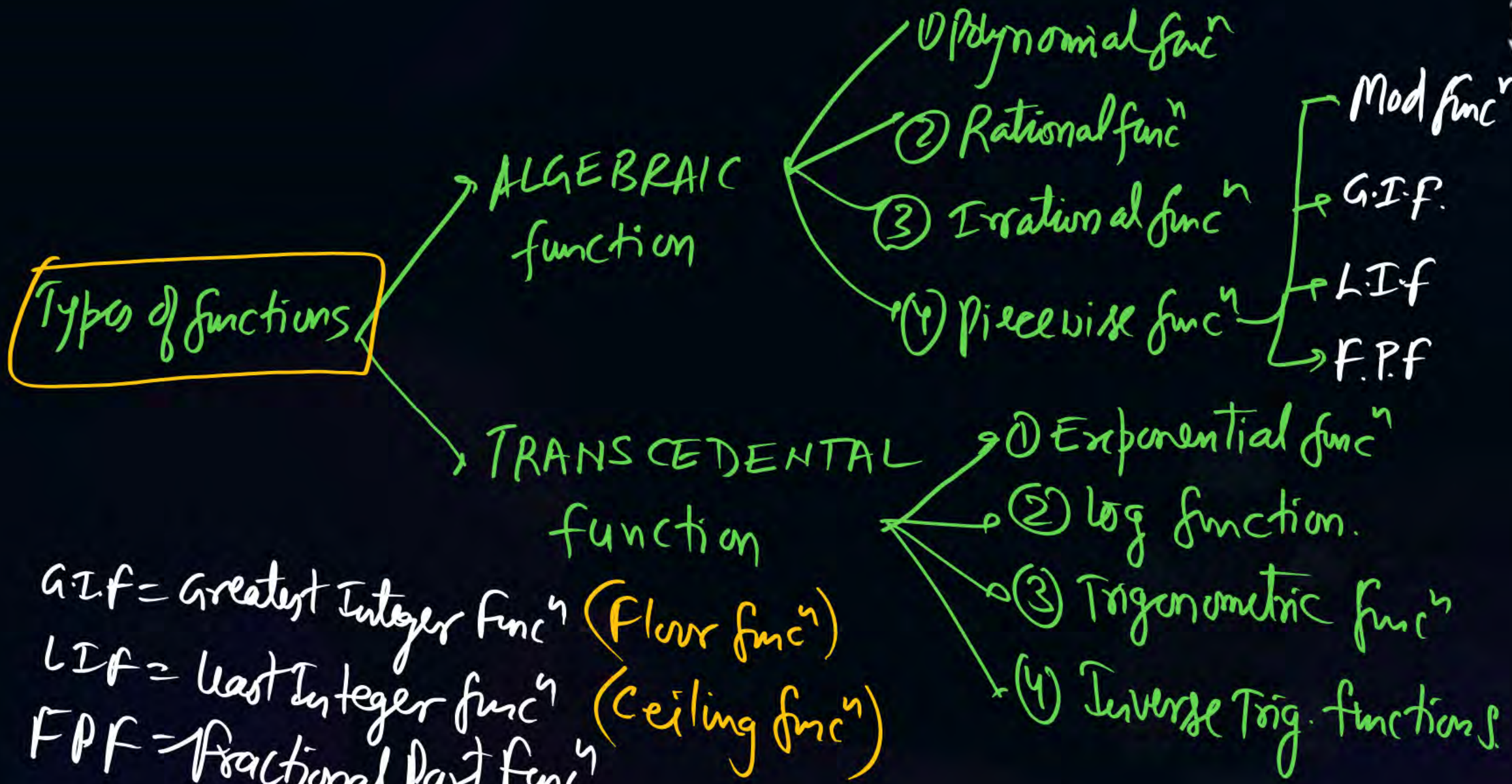


Periodic having $T=2\pi$

$f(x)=|\sin x|$



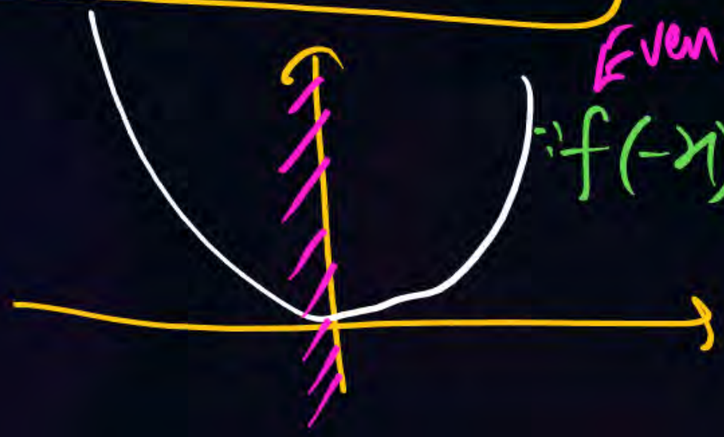
$T=\pi$ Periodic.


Types of functions
ALGEBRAIC function
① Polynomial func^n
② Rational func^n
③ Irrational func^n
④ Piecewise func^n
Mod func^n
G.I.f.
L.I.f
F.P.f
TRANSCEDENTAL function
① Exponential func^n
② log function.
③ Trigonometric func^n
④ Inverse Trig. functions
G.I.f = Greatest Integer Func^n (Floor func^n)
L.I.f = Least Integer func^n (Ceiling func^n)
F.P.F = Fractional Part Func^n.

CALCULUS

Function : $y = f(x) = x^2$

x :	1	2	3	4	5	6
y :	1	4	9	16	25	36



Even func^n

$\therefore f(-x) = (-x)^2$

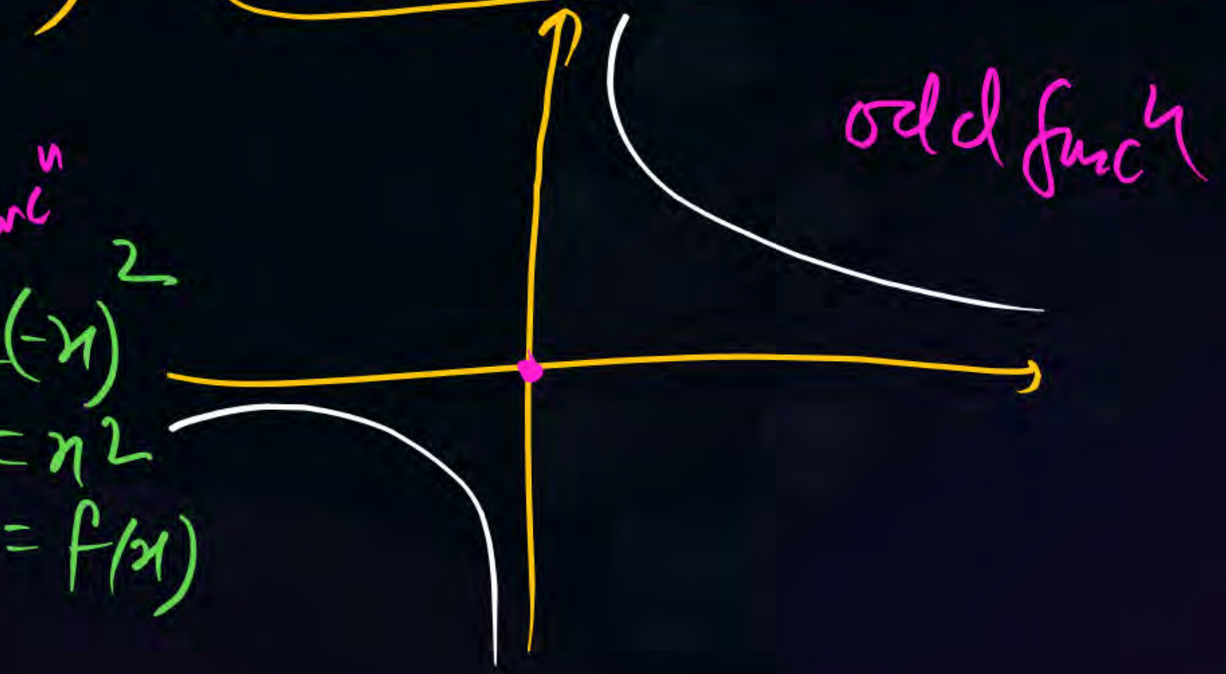
$= x^2$

$= f(x)$

$f(x) = y = \frac{1}{x}$ $\quad \therefore f(-x) = \frac{1}{(-x)}$

$= -\left(\frac{1}{x}\right) = -f(x)$

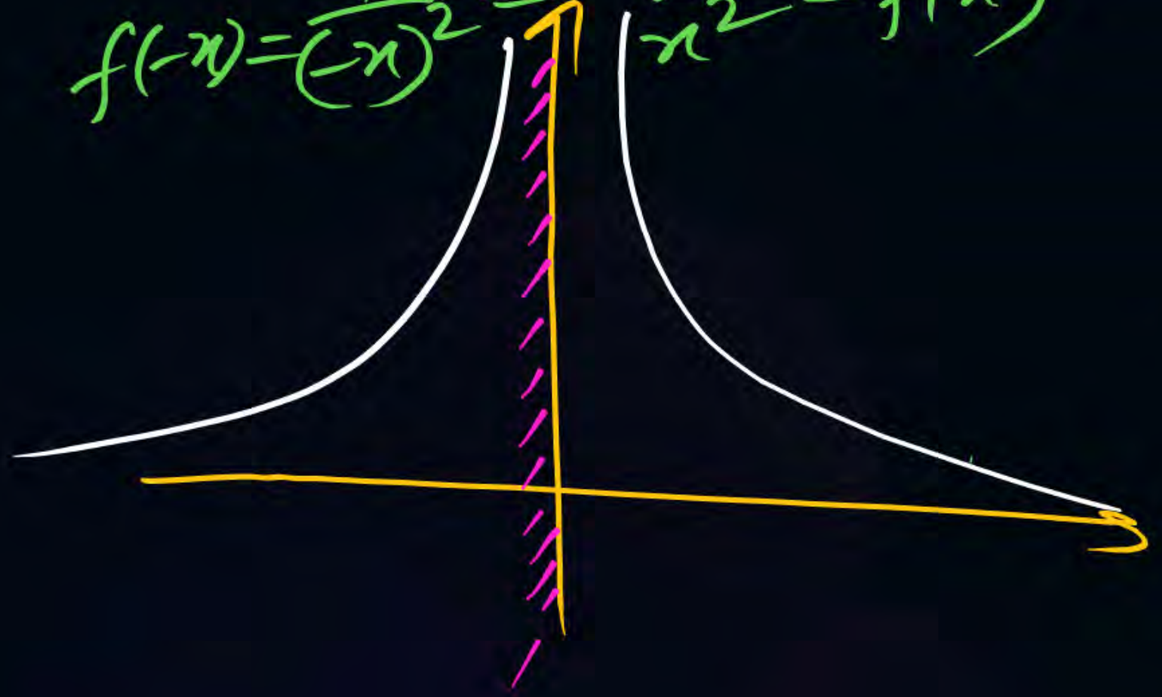
x :	1	2	3	4	5	6	7	8
y :	1	1/2	1/3	1/4	1/5	1/6	1/7	1/8



odd func^n

$f(x) = y = \frac{1}{x^2}$ Even func^n

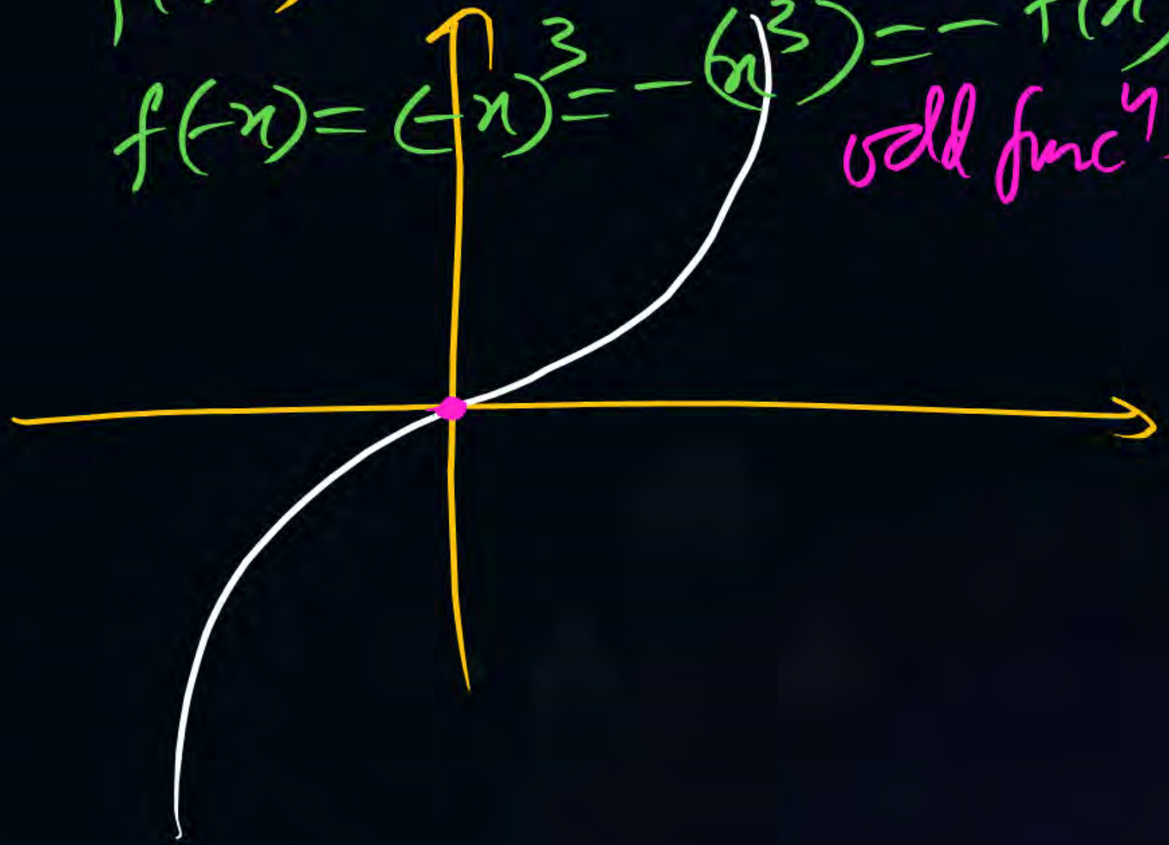
$f(-x) = \frac{1}{(-x)^2} = \frac{1}{x^2} = f(x)$



$f(x) = y = x^3$

$f(-x) = (-x)^3 = -(x^3) = -f(x)$

odd func^y.



$f(x) = y = e^x$

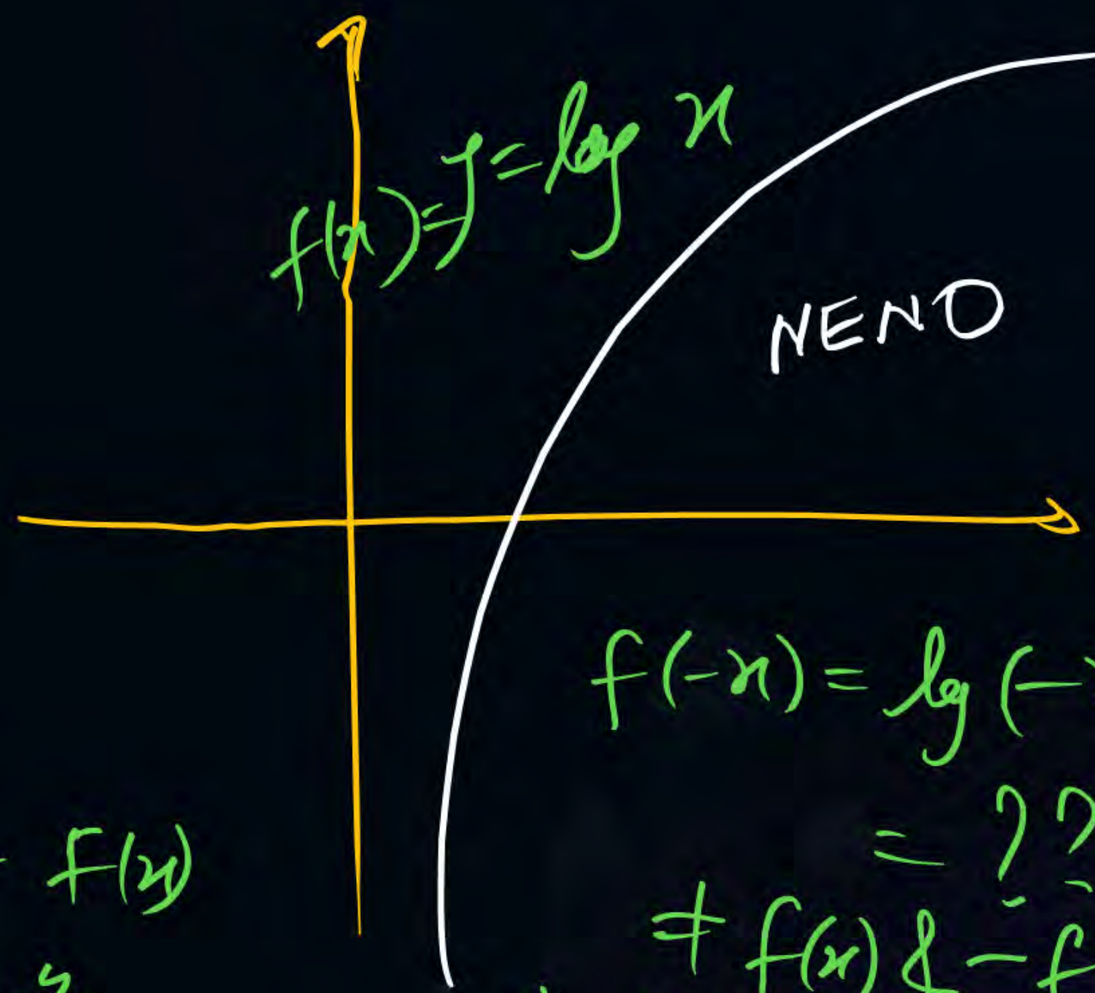


$\because f(-x) = e^{-x} = \frac{1}{e^x} = \frac{1}{f(x)}$

$\because f(-x) \neq +f(x)$ & $-f(x)$

$\therefore f(x)$ is NENO func^n

$f(x) = y = \log x$



$f(-x) = \log(-x)$

$= ??$

$\neq f(x)$ & $-f(x)$

$\therefore f(x)$ is NENO func^n

THANK - YOU

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